

Grab Pose Studio v1.0

Simulation System SDK — Internal Developer Guide

Overview

Grab Pose Studio is an in-house SDK tool that allows a single developer to record controller grab poses directly inside VR. It replaces the previous two-person workflow by moving the XR Rig into a dedicated recording studio during Play Mode. Recorded poses are saved to [Assets/RecordedPoses/](#) and work with the existing Grab Pose bake pipeline.

This is Grab Pose Studio v1. It covers controller pose recording only.

Before You Start

Before launching **Grab Pose Studio**, ensure the following:

-  **Quest Link** is connected and the headset is detected by Unity.
-  **Controllers are active** (not Hand Tracking). If needed, pick up both controllers to switch input mode.
-  **GrabPoseStudio.unity** has been added to **Build Settings**.
-  Open the **simulation scene** containing the objects you want to record.
-  Make sure the target objects have **GrabInteraction** components and are active in the scene.

Note: Grab Pose Studio works only in **Play Mode**. Launch it from **Tools** → **Grab Pose Studio** → **Launch Studio**.

 **Tip:** If you're using Grab Pose Studio for the first time, record one object completely (Left + Right pose), exit Play Mode, assign the generated [RecordedPose](#) assets using the existing **Grab Pose Recorder** workflow, and verify the grab before recording the remaining objects.

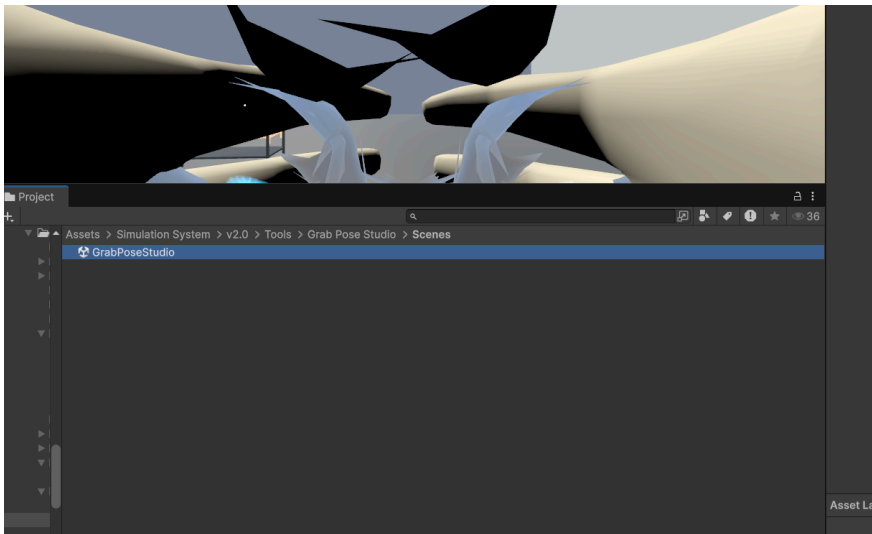
Features

- One-click launch from Edit Mode (**Tools** → **Grab Pose Studio** → **Launch Studio**)
 - Automatic XR Rig reposition into the Studio space
 - Auto-discovery of all runtime **GrabInteraction** objects
 - Next / Previous navigation through the full object list
 - Live Left / Right pose status per object
 - Full compatibility with the existing Grab Pose Recorder assignment and bake pipeline
 - BOT auto-hidden during session, auto-restored on Play Mode exit
 - No modifications to any simulation scene asset
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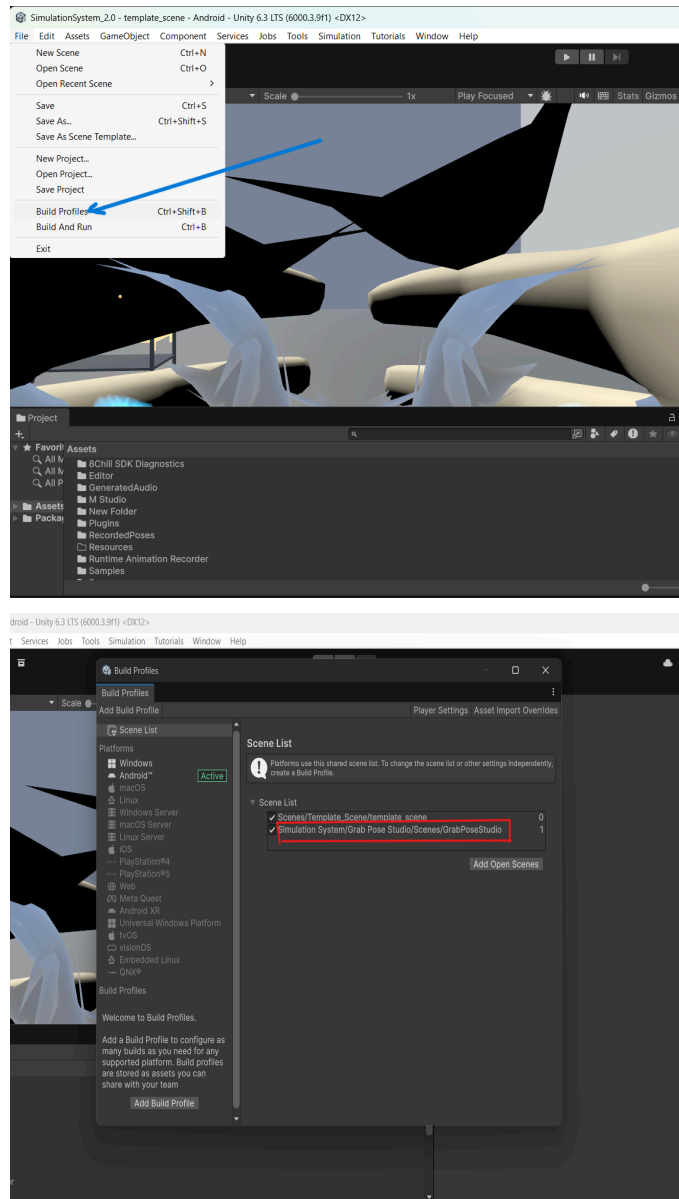
Setup

Already included in the SDK

1. Open **File** → **Build Settings**
2. Add **Assets/Simulation System/v2.0/Tools/Grab Pose Studio/Scenes/GrabPoseStudio.unity** to the scene list



3. Save



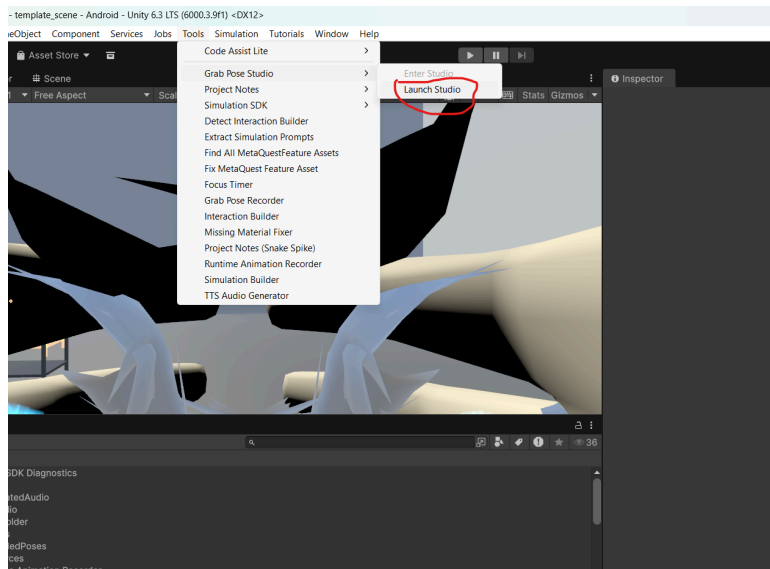
The scene must appear in Build Settings for **SceneManager** to load it at runtime. It does not need to be enabled for shipping builds.

Standalone package

1. Import the **GrabPoseStudio** package into your project
 2. Add **GrabPoseStudio.unity** to Build Settings as above
 3. Save
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Workflow

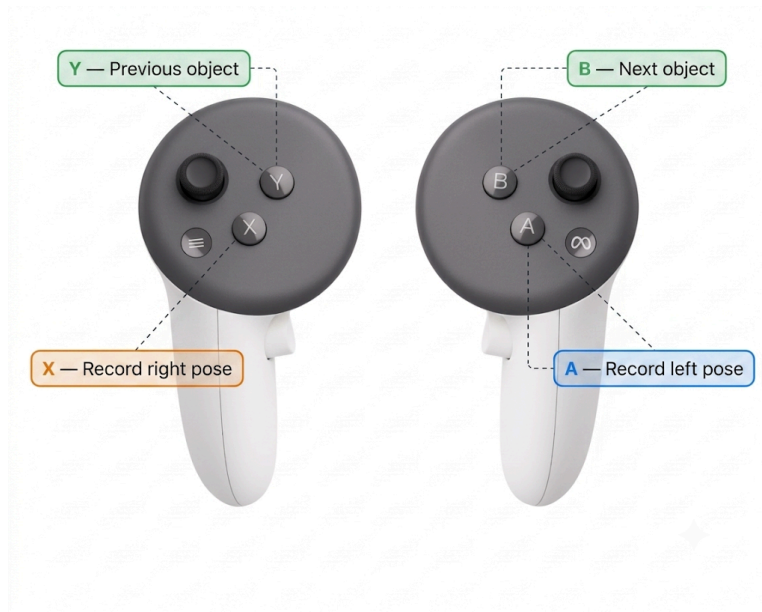
1. Open your simulation scene
2. Connect Quest Link and confirm controllers are active
3. Select **Tools** → **Grab Pose Studio** → **Launch Studio**
4. Unity enters Play Mode — the Studio loads, the XR Rig moves to **PlayerPoint**, the first grabbable appears at the Recording Point
5. Position your controller around the object and press **A** or **X** to record
6. Press **B** / **Y** to navigate to the next or previous object
7. Repeat until all required objects are recorded
8. Exit Play Mode — assets persist to **Assets/RecordedPoses/**
9. Assign and bake via the existing **Grab Pose Recorder** window



Controls

Button	Controller	Action
A	Right	Record Right pose
X	Left	Record Left pose
B	Right	Next object
Y	Left	Previous object

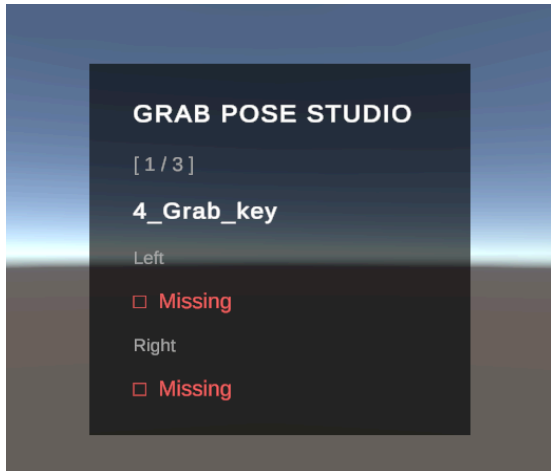
Record buttons are on the opposite controller to the hand being posed — the free hand presses Record while the other hand holds the pose.



Keyboard shortcuts (Editor only, Game view must have focus):

Key	Action
N	Next object
P	Previous object

Studio UI



Element	Description
Object name	<code>GameObject</code> name of the current grabbable — matches the generated asset filename prefix
Object index	Position in the discovered list, e.g. [3 / 14]
Left pose	☒ Recorded / ☒ Missing — checked live against <code>Assets/RecordedPoses/</code>
Right pose	☒ Recorded / ☒ Missing — checked live against <code>Assets/RecordedPoses/</code>
Feedback	2-second confirmation after each record action

Current Limitations

The following are **not** included in Grab Pose Studio v1 and are planned for a future release:

- Grab pose preview / visualization in VR
 - Automatic pose assignment to `GrabInteraction` components
 - Automatic baking
 - Exit Studio workflow (exit Play Mode manually)
 - Hand tracking recording
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Troubleshooting

Studio does not load `GrabPoseStudio.unity` is missing from Build Settings. Add it via **File** → **Build Settings** and save.

No grabbable objects found No active `GrabInteraction` components were found at launch. Confirm objects are enabled and the simulation scene is loaded before launching the Studio.

Recording has no effect / ❌ Recording Failed Confirm Quest Link is active and the headset is in controller mode (not hand tracking). Pick up both controllers to force the modality switch, then relaunch.

Version History

v1.1 (current)

- Build Settings guard — dialog appears if `GrabPoseStudio.unity` is missing, with one-click auto-add
- Assets/RecordedPoses/ folder created automatically on first recording
- Updated scene path: Assets/Simulation System/v2.0/Tools/Grab Pose Studio/
- Duplicate Build Settings entry protection
- World Space status panel replaces Game View IMGUI overlay
- BOT auto-hidden on Studio enter

v1.0

- Initial release
- One-click Launch Studio from Edit Mode
- Automatic XR Rig repositioning
- Runtime `GrabInteraction` discovery
- Next / Previous navigation with transform restore
- Left and Right controller pose recording
- RecordedPose asset generation to Assets/RecordedPoses/
- Live pose status per object